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WIND TURBINE BLADE AND METHOD FOR MANUFACTURING A WIND TURBINE BLADE

Abstract

A wind turbine blade is provided, including: a first and a second blade component connected with each other in an overlap region by thermal welding, a resistive element arranged between the first and second blade components in the overlap region as a remnant of the thermal welding, the resistive element having a first and a second terminal, a lightning conductor electrically connected with each of the first and second terminals, and a surge protection device, wherein one of the first and second terminals is electrically connected with the lightning conductor via the surge protection device. The resistive element which was used for the welding process and is left in the blade as a remnant of the welding process can be integrated into the lightning protection system of the blade.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a national stage of PCT Application No. PCT/EP2023/058539, having a filing date of Mar. 31, 2023, which claims priority to EP application Ser. No. 22/382,412.9, having a filing date of Apr. 29, 2022, the entire contents both of which are hereby incorporated by reference.

FIELD OF TECHNOLOGY

[0002] The following relates to a wind turbine blade and method for manufacturing a wind turbine blade.

BACKGROUND

[0003] Wind turbine blades are commonly made from fiber-reinforced resin. To facilitate manufacture, individual blade components are often pre-manufactured and connected with each other afterwards. For example, a pressure side and a suction side shell of a blade are manufactured separately and are then connected with each other at the leading and trailing edges. Further, for example, a shear web is manufactured separately and is then connected to a pressure side and a suction side shell of a blade. Known methods of connecting pre-manufactured blade components with each other include adhesive processes as well as infusion of a dry fiber lay-up in a connection region with resin. Further, US 2020/0171768 A1 discloses thermal welding for joining blade components.

SUMMARY

[0004] An aspect relates to an improved wind turbine blade and an improved method for manufacturing a wind turbine blade.

[0005] Accordingly, a wind turbine blade is provided. The wind turbine blade comprises: [0006] a first and a second blade component connected with each other in an overlap region by thermal welding, [0007] a resistive element arranged between the first and second blade components in the overlap region as a remnant of the thermal welding, the resistive element having a first and a second terminal, [0008] a lightning conductor electrically connected with each of the first and second terminals, and [0009] a surge protection device, wherein one of the first and second terminals is electrically connected with the lightning conductor via the surge protection device. [0010] Connecting the first and second blade components by thermal welding provides a faster connection method compared to a laminate joint because curing of an epoxy resin is not required. Further, an advantage of thermal welding compared to a glue joint is that an adhesive which adds significant weight to the blade can be avoided.

[0011] Further, the resistive element which was used for the welding process and is left in the blade as a remnant of the welding process can be integrated into the lightning protection system of the blade. By connecting the resistive element at its first and second terminals to the lightning conductor, flashovers during lightning between the resistive element and the lightning conductor can be prevented. In embodiments, in the case of high voltages at the lightning conductor caused by a lightning strike of the blade, the surge protection device opens a pathway for the lightning current to flow via the resistive element. Hence, large voltage differences between the resistive element and the lightning conductor are prevented. Thus, uncontrolled current flows such as flashovers and arcing between the resistive element and the lightning conductor are avoided.

[0012] The wind turbine blade is part of a rotor of a wind turbine. The wind turbine is an apparatus to convert the wind's kinetic energy into electrical energy. The wind turbine comprises, for

example, a rotor having one or more of the blades connected each to a hub, a nacelle including a generator, and a tower holding, at its top end, the nacelle. The tower of the wind turbine may be connected via a transition piece to a foundation of the wind turbine, such as a monopile in the seabed or a concrete foundation.

[0013] The wind turbine blade, e.g., its root section, is fixedly or rotatably connected to the hub. Apart from a (cylindrical) root section, the wind turbine blade is formed aerodynamically. The wind turbine blade comprises, for example, a pressure side (upwind side) and a suction side (downwind side). The pressure side and the suction side are connected with each other at a leading edge and a trailing edge. The pressure and suction sides and the leading and trailing edges define an airfoil of the wind turbine blade.

[0014] The wind turbine blade has a shell made, for example, from a fiber-reinforced resin. The blade shell is, for example, manufactured by arranging fiber lay-up in a mold, infusing the fiber lay-up with resin, and curing the resin. The blade shell surrounds, for example, an interior cavity of the blade. The wind turbine blade further comprises, for example, one or more shear webs connecting the blade shells of the pressure side and the suction side in the interior cavity of the blade. The shear web provides shear strength to the blade.

[0015] A pressure side shell, suction side shell, the one or more shear webs and/or other blade components may be pre-manufactured separately. Then, the pre-manufactured blade components (e.g., a first and a second blade component) may be joined with each other, for example, by thermal welding.

[0016] Thermal welding includes, for example, thermal resin welding of weldable resin such as thermoplastic resin or weldable thermoset resin. Thermal welding is, for example, thermoplastic welding.

[0017] The resistive element is, in particular, an electrically conductive element. The resistive element can produce heat by a current that is applied to the resistive element and that is flowing between the first and second terminals of the resistive element. The resistive element is, in particular, a heating element. The heat produced by the resistive element depends on the electrical resistance of the resistive element and the amount of current and/or voltage applied to the resistive element.

[0018] The resistive element can be made of a metal. For example, the resistive element is made of copper. Alternatively, the resistive element can be made of carbon fibers.

[0019] The resistive element is arranged between the blade components such that it is sandwiched between the blade components in the overlap region.

[0020] The overlap region is a region in which a portion of the first blade component overlaps with a portion of the second blade component. The overlap region is a joining region of the first and second blade components.

[0021] During manufacture, a thermoplastic resin or weldable thermoset resin may be placed between the first and second blade components in the overlap region, e.g., in contact with a surface of the first and/or second blade component and/or the resistive element. When the resistive element is heated up by applying an electrical current through the resistive element, the thermoplastic resin or weldable thermoset resin melts and/or softens and joins the blade components.

[0022] The lightning conductor is a down conductor of a lightning protection system of the blade. The lightning conductor is, for example, arranged parallel to a longitudinal direction of the blade. In the manufactured state of the wind turbine, the lightning conductor is, for example, electrically connected at a first end thereof with a ground potential and is connected at a second end thereof with one or more air termination elements of the lightning protection system.

[0023] One of the first and second terminals being electrically connected with the lightning conductor via the surge protection device means that only one of the first and second terminals is electrically connected with the lightning conductor via the surge protection device. The other one of the first and second terminals is electrically connected with the lightning conductor without an

intercalated surge protection device.

[0024] The first terminal of the resistive element is, in particular, electrically connected with a first connection portion of the lightning conductor. Further, the second terminal of the resistive element is electrically connected with a second connection portion of the lightning conductor, the first and second connection portions being separated from each other.

[0025] According to an embodiment, the surge protection device is configured to: [0026] provide a closed electrical connection between the respective terminal of the resistive element and the lightning conductor for lightning protection when a voltage applied to the surge protection device exceeds a threshold voltage, and [0027] provide an electrical isolation between the respective terminal of the resistive element and the lightning conductor for thermal welding of the first and second blade components when the applied voltage is at or below the threshold voltage.

[0028] Thus, during manufacture of the blade, the electrical connection of the resistive element with the lightning conductor for lightning protection can be already established, in particular before the thermal welding process. Nevertheless, the resistive element can still be used during manufacture as a heating element for the thermal welding process.

[0029] A voltage applied to the surge protection device during a lightning strike usually exceeds the threshold voltage such that the surge protection device acts as electrical connector.

[0030] Further, a voltage applied to the surge protection device during the thermal welding process is usually much lower than the threshold voltage such that the surge protection device acts as electrical isolator.

[0031] The lightning conductor is, for example, arranged in the interior cavity of the blade shell. The lightning conductor is, for example, arranged at the first and/or second blade component and/or at a shear web of the blade.

[0032] According to an embodiment, only one of the first and second terminals is connected to the lightning conductor via a surge protection device and the other one of the first and second terminals is connected to the lightning conductor by an electrical wire.

[0033] In embodiments, the other one of the first and second terminals is connected to the lightning conductor by only an electrical wire, i.e., without an intercalated surge protection device.

[0034] According to an embodiment, the resistive element is an elongated element extending in a longitudinal direction and having a first and a second end with respect to its longitudinal direction, and the first terminal is arranged at the first end and the second terminal is arranged at the second end.

[0035] Having an elongated resistive element (i.e., an elongated conductive element) in the blade bears the risk of large voltage difference between the first and second ends of the resistive element during lightning conditions. By connecting the first and second ends of the elongated resistive element with the lightning conductor, flashovers between the resistive element and the lightning conductor can be better prevented.

[0036] The length of the resistive element in the longitudinal direction is, for example, 3 meter or more, 5 meter or more, 7 meter or more and/or 10 meter or more.

[0037] According to an embodiment, the overlap region of the first and second blade components is an elongated region arranged (i.e., extending) parallel to a longitudinal direction of the blade and/or the resistive element is an elongated element arranged (i.e., extending) parallel to the longitudinal direction of the blade.

[0038] The elongated resistive element is, for example, arranged such that it is extending parallel to the lightning conductor.

[0039] According to an embodiment, the wind turbine blade comprises two or more of the resistive elements electrically connected to each other in series at their respective first and/or second terminals.

[0040] By using more than one resistive element, a joint between the first and second blade components over a large bonding line can be realized by thermal welding. Further, by using more

than one resistive element, the voltage required for heating a respective one of resistive elements can be kept small (e.g., at or below 690 V and/or 220 V).

[0041] The wind turbine blade may also comprise more than one surge protection device.

[0042] According to an embodiment, each of two neighboring ones of the two or more resistive elements are electrically connected with each other at their neighboring terminals, and the neighboring terminals are electrically connected with the lightning conductor by a common conductive wire and/or by a common surge protection device.

[0043] Thus, the number of conductive wires, conductive connections and/or surge protection devices can be reduced.

[0044] According to an embodiment, the wind turbine blade comprises an even number n of the resistive elements and a number p of the surge protection devices, wherein p is an even number fulfilling the equation $p=(n/2)$ or p is an odd number fulfilling the equation $p=(n/2)+1$.

[0045] In an embodiment, p is fulfilling the equation $p=(n/2)$ because in this case a smaller number of surge protection devices is necessary.

[0046] According to an embodiment, the wind turbine blade comprises an odd number m of the resistive elements and a number q of the surge protection devices, wherein q is an even number fulfilling the equation $q=[(m-1)/2]+1$.

[0047] According to an embodiment, the first blade component and/or the second blade component comprises a blade half shell, a lower blade shell, an upper blade shell, a pressure side shell, a suction side shell, a structural element, a reinforcement beam, a web and/or a shear web.

[0048] According to an embodiment, the threshold voltage is 1.5 kV or larger, 2 kV or larger, 2.5 kV or larger, 3 kV or larger, 10 kV or larger and/or 30 kV or larger.

[0049] According to a further aspect, a method for manufacturing a wind turbine blade is provided. In embodiments, the method comprises: [0050] a) arranging a first and a second blade component such that they overlap with each other in an overlap region, wherein a resistive element is arranged between the first and second blade components in the overlap region, the resistive element has a first and a second terminal, a lightning conductor is arranged such that it is electrically connected with each of the first and second terminals, and the lightning conductor is electrically connected to one of the first and second terminals via a surge protection device, and [0051] b) connecting the first and second blade components to each other by thermal welding including supplying an electrical current between the first and second terminals of the resistive element for heating the resistive element.

[0052] In step a), a weldable resin such as thermoplastic or weldable thermoset resin may be arranged between the first and second blade components.

[0053] According to an embodiment of the further aspect, a voltage applied at the surge protection device in step b) is at or below a threshold voltage such that: [0054] the surge protection device provides an electrical isolation between the one of the first and second terminals of the resistive element and the lightning conductor, and [0055] the electrical current for heating the resistive element is supplied through the electrical connection between the other one of the first and second terminals of the resistive element and the lightning conductor.

[0056] According to an embodiment of the further aspect, each of the first and second terminals are electrically connected in step b) with a power source for supplying the electrical current between the first and second terminals.

[0057] According to an embodiment of the further aspect, the method comprises the step of removing after step b) the electrical connection between each of the first and second terminals and the power source.

[0058] The embodiments and features described with reference to the wind turbine blade of the present invention apply mutatis mutandis to the method of the present invention.

[0059] Further possible implementations or alternative solutions of embodiments of the invention also encompass combinations—that are not explicitly mentioned herein—of features described

above or below with regard to the embodiments. The person skilled in the conventional art may also add individual or isolated aspects and features to the most basic form of embodiments of the invention.

Description

BRIEF DESCRIPTION

[0060] Some of the embodiments will be described in detail, with references to the following Figures, wherein like designations denote like members, wherein:

[0061] FIG. 1 shows a wind turbine according to an embodiment;

[0062] FIG. 2 shows a cross-section view of a wind turbine blade of the wind turbine of FIG. 1 according to an embodiment;

[0063] FIG. 3 shows a detail view of portion III of FIG. 2;

[0064] FIG. 4 illustrates electrical connections of a resistive element of the blade of FIG. 2 during thermal welding of blade components of the blade;

[0065] FIG. 5 shows a view similar as FIG. 4 but for an embodiment of a blade comprising two resistive elements;

[0066] FIG. 6 shows a view similar as FIG. 4 but for an embodiment of a blade comprising four resistive elements;

[0067] FIG. 7 shows a view similar as FIG. 5 but for a variant of the electrical connection;

[0068] FIG. 8 shows a view similar as FIG. 6 but for a variant of the electrical connection;

[0069] FIG. 9 shows a view similar as FIG. 4 but for an embodiment of a blade comprising three resistive elements;

[0070] FIG. 10 shows a view similar as FIG. 9 but for a variant of the electrical connection; and

[0071] FIG. 11 shows a flowchart illustrating a method for manufacturing a wind turbine blade of the wind turbine of FIG. 1 according to an embodiment.

DETAILED DESCRIPTION

[0072] FIG. 1 shows a wind turbine 1 according to an embodiment. The wind turbine 1 comprises a rotor 2 having one or more blades 3 connected to a hub 4. The hub 4 is connected to a generator (not shown) arranged inside a nacelle 5. During operation of the wind turbine 1, the blades 3 are driven by wind to rotate and the wind's kinetic energy is converted into electrical energy by the generator in the nacelle 5. The nacelle 5 is arranged at the upper end of a tower 6 of the wind turbine 1. The tower 6 is erected on a foundation 7 such as a monopile or concrete foundation. The foundation 7 is connected to and/or driven into the ground or seabed.

[0073] FIG. 2 shows a cross-section view of a wind turbine blade 3 of the wind turbine 1 of FIG. 1.

[0074] The blade 3 comprises a blade shell 8 including a lower blade shell 9 and an upper blade shell 10. The lower and upper blade shells 9, 10 are lower and upper blade shells with respect to a manufacturing position. The lower blade shell 9 is, for example, a pressure side shell and the upper blade shell 10 is, for example, a suction side shell or vice versa. The blade shell 8 surrounds an interior cavity 11 of the blade 3.

[0075] The blade 3 further comprises one or more structural elements 12 running in a longitudinal direction L1 of the blade 3. The longitudinal direction L1 of the blade 3 is, for example, pointing from a root of the blade 3 to a tip of the blade 3. The structural element 12 comprises fiber composite material, in particular glass fiber mats. The structural element 12 can be a shear web, a spar cap or the like. In FIG. 2, a shear web 13 is shown as an example for a structural element 12. Although not shown in the figures, the blade 3 can also comprise more than one shear web 13. The shear web 13 connects the blade shells 9, 10 of the pressure side and the suction side in the interior cavity 11 of the blade 3 and is providing shear strength to the blade 3. The shear web 13 comprises two flanges 14, 15 that are attached to inner surfaces 16, 17 of the shells 9, 10, respectively.

[0076] The blade **3** is, for example, assembled from several pre-manufactured blade components such as the lower blade shell **9**, the upper blade shell **10**, the structural element **12** and/or the shear web **13**. Such pre-manufactured blade components may, for example, be connected to each other at joints **18**, **19**, **20**, **21** by thermal welding. For example, the lower blade shell **9** is connected to the upper blade shell **10** in a first and a second joint **18**, **19** by thermal welding. The first and second connection regions **18**, **19** are, in particular, arranged at a leading edge **22** and a trailing edge **23** of the blade **3**. Furthermore, the structural element **12** such as the shear web **13** is connected to the lower blade shell **9** in a third joint **20** by thermal welding and/or is connected to the upper blade shell **10** in a fourth joint **21** by thermal welding.

[0077] In the following, exemplarily, the fourth joint **21** between the shear web **13** and the upper blade shell **10** by thermal welding is described. However, the following description may also be applied to one, more or all of the other joints **18**, **19**, **20**.

[0078] FIG. **3** shows a detailed view of portion III of FIG. **2**. Visible in FIG. **3** is a portion of the shear web **13** including its upper flange **15** and a portion of the upper blade shell **10**. The shear web **13** is an example of a first blade component. Further, the upper blade shell **10** is an example of a second blade component. The shear web **13** and the upper blade shell **10** are connected with each other in an overlap region **24** by thermal welding. For providing the necessary heat for thermal welding, the blade **3** comprises a resistive element **25**. The resistive element **25** is arranged between the shear web **13** (in particular the upper flange **15** of the shear web **13**) and the upper blade shell **10** in the overlap region **24** during manufacture. Further, as can be seen in FIG. **3**, the resistive element **25** remains after the manufacturing process in the blade **3**. Further indicated in FIG. **3** is that during the manufacturing process a weldable resin **26** is arranged between the shear web **13** (in particular the upper flange **15** of the shear web **13**) and the upper blade shell **10** in the overlap region **24**.

[0079] The resistive element **25** which is used as a heating element for the welding process during manufacture is left in the blade **3** as a remnant of the welding process. Further, the resistive element **25** is electrically connected (reference sign **27**) to a lightning conductor **28** and is, thus, integrated into a lightning protection system (not shown) of the blade **3**.

[0080] FIG. **4** illustrates electrical connections of the resistive element **25** during thermal welding of blade components, i.e., of the shear web **13** and the upper blade shell **10**. FIG. **4** shows a cross-section along line A-A in FIG. **2**.

[0081] The resistive element **25** is an elongated element extending in a longitudinal direction **L2** and parallel to the longitudinal direction **L1** of the blade **3**. The resistive element **25** has a first end **29** and a second end **30** with respect to the longitudinal directions **L1**, **L2**.

[0082] Further, the resistive element **25** has first and second terminals **31**, **32** for electrical connection. In embodiments, the first terminal **31** is arranged at the first end **29** of the resistive element **25**. Further, the second terminal **32** is arranged at the second end **30** of the resistive element **25**.

[0083] In order to prevent flashovers during operation of the wind turbine **1** between the resistive element **25** remaining in the blade **3** after manufacture and the lightning conductor **28**, the first and second terminals **31**, **32** of the resistive element **25** are electrically connected to the lightning conductor **28**. In embodiments, one of the first and second terminals **31**, **32** of the resistive element **25** is electrically connected to the lightning conductor **28** by an electrical wire **33**. Further, the other one of the first and second terminals **31**, **32** of the resistive element **25** is electrically connected to the lightning conductor **28** via a surge protection device **34** and a further electrical wire **35**. In the example of FIG. **4**, the first terminal **31** is electrically connected to the lightning conductor **28** by the electrical wire **33**, and the second terminal **32** is electrically connected to the lightning conductor **28** via the surge protection device **34**.

[0084] The surge protection device **34** is configured to provide a closed electrical connection between the second terminal **32** of the resistive element **25** and the lightning conductor **28** when a

voltage V applied to the surge protection device **34** exceeds a threshold voltage $V_{sub.th}$. Since a voltage V applied to the surge protection device **34** during a lightning strike usually exceeds the threshold voltage $V_{sub.th}$, the surge protection device **34** acts as electrical connector in this case. [0085] Further, the surge protection device **34** is configured to provide an electrical isolation between the second terminal **32** of the resistive element **25** and the lightning conductor **28** when the applied voltage V is at or below the threshold voltage $V_{sub.th}$.

[0086] In this manner, during manufacture of the blade **3**, the electrical connection (**33**, **34**, **35**) of the resistive element **25** with the lightning conductor **28** for lightning protection can be already established (in particular before the thermal welding process). Nevertheless, the resistive element **25** can still be used during manufacture as a heating element for the thermal welding process.

[0087] FIG. **4** shows the electrical connections of the resistive element **25** during the welding process. Each of the first and second terminals **31**, **32** is electrically connected with a power source **36** for supplying electrical current I between the first and second terminals **31**, **32** for heating the resistive element **25**. In embodiments, the first terminal **31** is electrically connected by the electrical wire **33** to the lightning conductor **28** and the lightning conductor **28** is electrically connected by a further electrical wire **37** to the power source **36**. Further, the second terminal **32** is electrically connected by a further electrical wire **38** to the power source **36**. Since the voltage V applied to the surge protection device **34** during the thermal welding process is lower than the threshold voltage $V_{sub.th}$, the surge protection device **34** acts as an electrical isolator in this case.

[0088] After the thermal welding process, the electrical connection between each of the first and second terminals **31**, **32** and the power source **36** are removed. For example, the electrical wires **37** and **38** are disconnected from the first and second terminals **31**, **32**, respectively.

[0089] The blade **3** may also comprise two or more of the resistive elements **25** electrically connected to each other in series, as shown in FIGS. **5** to **10**. Using more than one resistive element **25** allows an easier heating of the resistive element even when the thermal welding process is carried out for a long bondline. Specially, since a length of the blade **3** can be as large as 100 meter and more, using several resistive elements **25** is desired.

[0090] As shown in FIGS. **5** to **10**, different electrical connections of the resistive element **25** are possible depending on the number n , m of resistive elements **25** connected in series and the number p , q of applied surge protection devices **34**.

[0091] FIGS. **5** to **8** show examples with an even number n of resistive elements **25**. FIGS. **9** and **10** show examples with an odd number m of resistive elements **25**.

[0092] FIG. **5** shows an example of two resistive elements **125a**, **125b** connected with each other in series at their neighboring terminals **132a**, **131b**.

[0093] For lightning protection during operation of the wind turbine **1**, a first resistive element **125a** is electrically connected at its first terminal **131a** via an electrical wire **133a** to the lightning conductor **28**. Further, the neighboring terminals **132a**, **131b** of the first and second resistive elements **125a**, **125b** (i.e., the second terminal **132a** of the first resistive element **125a** and the first terminal **131b** of the second resistive element **125b**) are electrically connected via a common surge protection device **134** to the lightning conductor **28**. Furthermore, the second resistive element **125b** is electrically connected at its second terminal **132b** via an electrical wire **133b** to the lightning conductor **28**. These electrical connections are, in particular, established before the thermal welding process.

[0094] For thermal welding, additionally, the lightning conductor **28** is electrically connected by an electrical wire **137** to the power source **36**. Further, the neighboring terminals **132a**, **131b** of the first and second resistive elements **125a**, **125b** are connected by a common electrical wire **138** to the power source **36**.

[0095] The surge protection device **134** acts as isolator for the voltage applied by the power source **36** (e.g., less than 690 V). Hence, an electrical current I is flowing from the power source **36** via the electrical wire **137**, the lightning conductor **28**, the electrical wire **133a**, the first resistive element

125a and the electrical wire **138** back to the power source **38**. Thereby, the first resistive element **125a** is heated.

[0096] Furthermore, an electrical current **I** is flowing from the power source **36** via the electrical wire **137**, the lightning conductor **28**, the electrical wire **133b**, the second resistive element **125b** and the electrical wire **138** back to the power source **38**. Thereby, the second resistive element **125b** is heated.

[0097] In embodiments, for two resistive elements **125a**, **125b** only one surge protection device **134** is required.

[0098] FIG. **6** shows an example of four resistive elements **225a**, **225b**, **225c**, **225d** connected with each other in series at their respective terminals. In this case two surge protection devices **234a**, **234b** and three electrical wires **233a**, **233b**, **233c** are used for electrically connecting the resistive elements **225a**, **225b**, **225c**, **225d** to the lightning conductor **28**.

[0099] In embodiments, for four resistive elements **225a**, **225b**, **225c**, **225d** only two surge protection devices **234a**, **234b** are required.

[0100] In embodiments, in the examples of FIGS. **5** and **6**, the number of resistive elements **125a**, **125b**, **225a-225d** is even and the number **p** of required surge protection devices **134**, **234a**, **234b** fulfills the equation $p=(n/2)$.

[0101] FIGS. **7** and **8** show examples for an even number **n** of resistive elements **325a**, **325b**, **425a-425d** for which the number **p** of required surge protection devices **334a**, **334b**, **434a-434c** is an odd number fulfilling the equation $p=(n/2)+1$.

[0102] In the example of FIG. **7**, for lightning protection during operation of the wind turbine **1**, a first resistive element **325a** is electrically connected at its first terminal **331a** via a surge protection device **334a** to the lightning conductor **28**. Further, the neighboring terminals **332a**, **331b** of the first and second resistive elements **325a**, **325b** are electrically connected via a common electrical wire **333** to the lightning conductor **28**. Furthermore, the second resistive element **325b** is electrically connected at its second terminal **332b** via another surge protection device **334b** to the lightning conductor **28**. These electrical connections are, in particular, established before the thermal welding process.

[0103] For thermal welding, additionally, the power source **36** is connected by an electrical wire **337** to the lightning conductor **28**. Further, the first terminal **331a** of the first resistive element **325a** is electrically connected by an electrical wire **338a** to the power source **36**. Furthermore, the second terminal **332b** of the second resistive element **325b** is connected by an electrical wire **338b** to the power source **16**.

[0104] In comparison to the case of FIG. **5**, for the electrical connection of FIG. **7** two surge protection devices **334a**, **334b** are required for two resistive elements **325a**, **325b**.

[0105] FIG. **8** shows an example of four resistive elements **425a**, **425b**, **425c**, **425d** connected in series for which three surge protection devices **434a**, **434b**, **434c** and two electrical wires **433a**, **433b** are used for electrically connecting the resistive elements **425a**, **425b**, **425c**, **425d** with the lightning conductor **28**.

[0106] FIGS. **9** and **10** show examples with an odd number **m** of resistive elements **525a-525c** and **625a-625c**. In this case, the number **q** of required surge protection devices is an even number fulfilling the equation $q=[(m-1)/2]+1$.

[0107] In the example of FIG. **9**, for lightning protection, three resistive elements **525a**, **525b**, **525c** are electrically connected to the lightning conductor **28** using two surge protection devices **534a**, **534b** and two electrical wires **533a**, **533b**.

[0108] Furthermore, for thermal welding in addition three electrical wires **537**, **538a**, **538b** are used as illustrated in FIG. **9**.

[0109] The example of FIG. **10** is a variant of the example of FIG. **9** with a different arrangement of the surge protection device **634a**, **634b**. However, the number of used resistive elements **625a**, **625b**, **625c** and the number of used surge protection device **634a**, **634b** is the same as for FIG. **10**.

[0110] To conclude, the resistive elements which were used for the welding process and are left in the blade **3** as a remnant of the welding process are integrated into the lightning protection system of the blade **3**. Hence, flashovers during lightning between the resistive elements and the lightning conductor **28** can be prevented. In embodiments, in the case of high voltages at the lightning conductor **28** caused by a lightning strike of the blade **3**, the surge protection devices each open a pathway for the lightning current to flow via the resistive elements. Hence, large voltage differences between the resistive elements and the lightning conductor **28** are prevented.

[0111] Moreover, as the electrical connection between the resistive elements and the lightning conductor **28** are installed before the welding process for joining the blade components, they can be easier installed.

[0112] In the following, a method for manufacturing a wind turbine blade is described with reference to FIG. **11**.

[0113] In a first step **S1** of embodiments of the method a first and a second blade component (for example, the shear web **13** and the upper shell **10**, FIG. **2**) are arranged such that they overlap with each other in an overlap region **24** (FIG. **3**). Further, a resistive element **25** is arranged between the first and second blade components **10**, **13** in the overlap region **24**. The resistive element **25** has a first and a second terminal **31**, **32** (FIG. **4**). Furthermore, a lightning conductor **28** is arranged such that it is electrically connected with each of the first and second terminals **31**, **32**, wherein the lightning conductor **28** is electrically connected to one of the first and second terminals **31**, **32** via a surge protection device **34**.

[0114] In a second step **S2** of embodiments of the method, the first and second blade components **10**, **13** are connected to each other by thermal welding. Thermal welding includes supplying an electrical current **I** between the first and second terminals **31**, **32** of the resistive element **25** for heating the resistive element **25**.

[0115] In embodiments, each of the first and second terminals **31**, **32** is electrically connected with a power source **26** for supplying the electrical current **I** between the first and second terminals **31**, **32**. Further, a voltage applied by the power supply (e.g., 690 V or less) is at or below a threshold voltage $V_{sub.th}$ of the surge protection device **34** (e.g., 2.5 kV or more). Hence, the surge protection device **34** provides an electrical isolation between the one of the first and second terminals **31**, **32** of the resistive element **25** and the lightning conductor **28**. In addition, the electrical current **I** for heating the resistive element **25** is supplied through the electrical connection between the other one of the first and second terminals **31**, **32** of the resistive element **25** and the lightning conductor **28**.

[0116] In a third step **S3** of embodiments of the method, the electrical connection between each of the first and second terminals **31**, **32** and the power source **36** is removed.

[0117] Although the present invention has been disclosed in the form of embodiments and variations thereon, it will be understood that numerous additional modifications and variations could be made thereto without departing from the scope of the invention.

[0118] For the sake of clarity, it is to be understood that the use of “a” or “an” throughout this application does not exclude a plurality, and “comprising” does not exclude other steps or elements.

Claims

1. A wind turbine blade, comprising: a first blade component and a second blade component connected with each other in an overlap region by thermal welding; a resistive element arranged between the first blade component and the second blade components in the overlap region as a remnant of the thermal welding, the resistive element having a first terminal and a second terminal; a lightning conductor electrically connected with each of the first terminal and the second terminal; and a surge protection device, wherein one of the first terminal and the second terminal is electrically connected with the lightning conductor via the surge protection device.

2. The wind turbine blade according to claim 1, wherein the surge protection device is configured to: provide a closed electrical connection between the respective terminal of the resistive element and the lightning conductor for lightning protection when a voltage applied to the surge protection device exceeds a threshold voltage, and provide an electrical isolation between the respective terminal of the resistive element and the lightning conductor for thermal welding of the first blade component and the second blade components when the applied voltage is at or below the threshold voltage.
3. The wind turbine blade according to claim 1, wherein only one of the first terminal and the second terminal is connected to the lightning conductor via a surge protection device and the other one of the first terminal and the second terminal is connected to the lightning conductor by an electrical wire.
4. The wind turbine blade according to claim 1, wherein the resistive element is an elongated element extending in a longitudinal direction and having a first end and a second end with respect to the longitudinal direction, and the first terminal is arranged at the first end and the second terminal is arranged at the second end.
5. The wind turbine blade according to claim 1, wherein the overlap region of the first blade component and the second blade component is an elongated region arranged parallel to a longitudinal direction of the wind turbine blade and/or the resistive element is an elongated element arranged parallel to the longitudinal direction of the wind turbine blade.
6. The wind turbine blade according to claim 1, comprising two or more of the resistive elements electrically connected to each other in series at respective first terminal and/or second terminal.
7. The wind turbine blade according to claim 6, wherein each of two neighboring ones of the two or more resistive elements are electrically connected with each other at neighboring terminals, and the neighboring terminals are electrically connected with the lightning conductor by a common conductive wire and/or by a common surge protection device.
8. The wind turbine blade according to claim 6, comprising an even number n of the resistive elements, and a number p of the surge protection devices, wherein p is an even number fulfilling the equation $p=(n/2)$ or p is an odd number fulfilling the equation $p=(n/2)+1$.
9. The wind turbine blade according to claim 6, comprising an odd number m of the resistive elements and a number q of the surge protection devices, wherein q is an even number fulfilling the equation $q=[(m-1)/2]+1$.
10. The wind turbine blade according to claim 1, wherein the first blade component and/or the second blade component comprises a blade half shell, a lower blade shell, an upper blade shell, a pressure side shell, a suction side shell, a structural element, a reinforcement beam, a web and/or a shear web.
11. The wind turbine blade according to claim 2, wherein the threshold voltage is 1.5 kV or larger, 2 kV or larger, 2.5 kV or larger, 3 kV or larger, 10 kV or larger, and/or 30 kV or larger.
12. A method for manufacturing a wind turbine blade, comprising: a) arranging first blade component and a second blade component such that the first blade component and the second blade component overlap with each other in an overlap region, wherein a resistive element is arranged between the first blade component and the second blade components in the overlap region, the resistive element has a first terminal and a second terminal, a lightning conductor is arranged such that the lightning conductor is electrically connected with each of the first terminal and the second terminal, and the lightning conductor is electrically connected to one of the first terminal and the second terminal via a surge protection device; and b) connecting the first blade component and the second blade components to each other by thermal welding including supplying an electrical current between the first terminal and the second terminal of the resistive element for heating the resistive element.
13. The method according to claim 12, wherein a voltage applied at the surge protection device in step b) is at or below a threshold voltage such that: the surge protection device provides an

electrical isolation between the one of the first terminal and the second terminals of the resistive element and the lightning conductor, and the electrical current for heating the resistive element is supplied through the electrical connection between the other one of the first terminal and the second terminal of the resistive element and the lightning conductor.

14. The method according to claim 12, wherein each of the first terminal and the second terminals are electrically connected in step b) with a power source for supplying the electrical current AA-between the first terminal and second terminal.

15. The method according to claim 14, comprising the step of removing after step b) the electrical connection between each of the first terminal and the second terminals and the power source.
